QQ-R-566B
July 5, 1973
SUPERSEDING
Fed. Spec. QQ-R-566A
March 10, 1964

FEDERAL SPECIFICATION

RODS AND ELECTRODES, WELDING, ALUMINUM AND ALUMINUM ALLOYS

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers the requirements for drawn or cast bare aluminum and aluminum alloy welding rods and drawn electrodes for use with gas and arc welding processes.

1.2 Classification. Aluminum and aluminum alloy welding rods and electrodes shall be of the classes, by chemical composition, listed in table I, as specified (see 6.2).

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issues in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

Federal Specifications:

BB-H-1168	Helium, Technical.
QQ-A-250/1	Aluminum Alloy, 1100, Plate and Sheet.
QQ-A-250/2	Aluminum Alloy, 3003, Plate and Sheet.
QQ-A-250/6	Aluminum Alloy, 5083, Plate and Sheet.
QQ-A-250/7	Aluminum Alloy, 5086, Plate and Sheet.
QQ-A-250/8	Aluminum Alloy, 5052, Plate and Sheet.

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Federal Specifications:(Continued)

QQ-A-250/9	Aluminum Alloy, 5456, Plate and Sheet.
QQ-A-250/10	Aluminum Alloy, 5454, Plate and Sheet.
QQ-A-250/11	Aluminum Alloy, 6061, Plate and Sheet.
QQ-A-596	Aluminum Alloy, Permanent and Semipermanent Mold Castings.
QQ-A-601	Aluminum Alloy, Sand Casting.

Federal Standards :

Fed. Std. 123 - Marking for Domestic Shipment (Civil Agencies).

Fed. Test Method Std. No. 151 - Metals; Test Methods.

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications and Standards and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U. S. Government Printing Office, Washington, DC 20402.

(Single copies of this specification and other Federal Specifications required by activities outside the Federal Government for bidding purposes are available without charge from Business Service Centers at the General Services Administration Regional Offices in Boston, New York, Washington, DC, Atlanta, Chicago, Kansas City, MO, Fort Worth, Denver, San Francisco, Los Angeles, and Seattle, WA.

(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks and the Index of Federal Specifications and Standards from established distribution points in their agencies.)

Military Specifications:

MIL-I-6866	Inspection, Penetrant Method Of.
MIL-F-6939	Flux, Aluminum and Aluminum Alloy, Gas Welding.

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Military Specifications: (Continued)

MIL-A-8920	Aluminum Alloy, Plate and Sheet 2219.
MIL-W-10430	Welding Rods and Electrodes, Preparation For Delivery of.
MIL-A-18455	Argon, Technical.

Military Standards:

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129	Marking for Shipment and Storage.
MIL-STD-437	X-Ray Standards for Bare Aluminum Alloy Electrode Welds.
MIL-STD-453	Inspection, Radiographic.

(Copies of Military Specifications and Standards required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

American National Standards Institute (ANSI), Inc., Standard:

Z49.1	Safety in Welding and Cutting.

(Application for copies of ANSI standard should be addressed to either the American National Standards Institute, 1430 Broadway, New York, NY 10018 or the American Welding Society, 2501 N.W. 7th Street, Miami, FL 33125.)

Society of Automotive Engineers (SAE) Standards:

AMS 4000	Sheet and Plate - 99.6 Aluminum.
AMS 4222	Casting, Sand - 4 Cu, 2 Ni, 1.5 Mg, Solution Treated and Overaged.

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(Application for copies of AMS standards should be addressed to the Society of Automotive Engineers, 2 Pennsylvania Plaza, New York, NY 10001.)

American Society for Testing & Materials (ASTM) Standard:

E8	Standard Methods of Tension Testing of Metallic Materials.

(Application for copies of ASTM standards should be addressed to the American Society for Testing & Materials, 1916 Race Street, Philadelphia, PA 19103.)

3. REQUIREMENTS

3.1 Materials. The aluminum and aluminum alloy welding rods and electrodes shall be made by any process from such materials that will yield a product conforming with all the requirements of this specification.

3.1.1 Manufacture. Welding rods shall be drawn or cast, as specified. Welding electrodes shall be a drawn product (see 6.2).

3.2 Chemical Composition. The chemical composition of the welding rods and electrodes for each class shall be as specified in table I (see 6.2).

3.3 Form. Materials shall be furnished in straight lengths, in coils or wound on spools, as specified (see 6.2).

3.3.1 Size. Welding rods and electrodes shall be furnished in a diameter shown in table II, as specified (see 6.2). Individual determinations of diameters shall not vary from the specified nominal by more than + 0.001 inch and - 0.002 inch for drawn rods and electrodes or by more than ± 0.05 inch for cast round rods.

3.3.2 Length. Unless otherwise specified (see 6.2), drawn welding rods furnished in straight lengths shall be 36, +0, -1/2 inch long. Cast welding rods shall be furnished in straight lengths of 12, 18, 24, or 36 inches, as specified (see 6.2). Cast welding rods shall not vary more than ± 1/2 inch in length from that specified. Drawn welding rods or electrodes, furnished in coils or wound on spools, shall be one continuous length. No joints or butt splices shall be made in the finished diameter.

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TABLE I

Chemical Composition of Rods and Electrodes

Class	Elements (Percent) 1/														
	Cu	Mg	Mn	Cr	Si	Fe	Zn	Ti	Be	V	Zr	Ni	Others		Al
													Each	Total	
1100	0.05-0.20	-	0.05	-	2/	2/	0.10	-	0.0008	-	-	-	0.05	0.15	99.00 MIN
1260	0.04	-	0.01	-	3/	3/	-	-	0.0008	-	-	-	0.03	-	99.60 MIN
2319	5.8-6.8	0.02	0.20-0.40	-	0.20	0.30	0.10	0.10-0.20	0.0008	0.05-0.15	0.10-0.25	-	0.05	0.15	Remainder
4043	0.30	0.05	0.05	-	4.5-6.0	0.8	0.10	0.20	0.0008	-	-	-	0.05	0.15	Remainder
4047	0.30	0.10	0.15	-	11.0 13.0	0.8	0.20	-	0.0008	-	-	-	0.05	0.15	Remainder
4145	3.3-4.7	0.15	0.15	0.15	9.3-10.7	0.8	0.20	-	0.0008	-	-	-	0.05	0.15	Remainder
5039	0.03	3.3-4.3	0.30-0.50	0.10-0.20	0.10	0.40	2.4-3.2	0.10	0.0008	-	-	-	0.05	0.10	Remainder
5183	0.10	4.3-5.2	0.50-1.0	0.05-0.25	0.40	0.40	0.25	0.15	0.0008	-	-	-	0.05	0.15	Remainder
5356	0.10	4.5-5.5	0.05-0.20	0.05-0.20	5/	5/	0.10	0.06-0.20	0.0008	-	-	-	0.05	0.15	Remainder
5554	0.10	2.4-3.0	0.50-1.0	0.05-0.20	3/	3/	0.25	0.05-0.20	0.0008	-	-	-	0.05	0.15	Remainder
5556	0.10	4.7-5.5	0.50-1.0	0.05-0.20	3/	3/	0.25	0.05-0.20	0.0008	-	-	-	0.05	0.15	Remainder
5654	0.05	3.1-3.9	0.01	0.15-0.35	4/	4/	0.20	0.05-0.15	0.0008	-	-	-	0.05	0.15	Remainder
295.0	4.0-5.0	0.03	0.35	-	0.7-1.5	1.0	0.35	0.25	0.0008	-	-	-	0.05	0.15	Remainder
242.0	3.5-4.5	1.2-1.8	0.35	0.25	0.7	1.0	0.35	0.25	0.0008	-	-	1.7-2.3	0.05	0.15	Remainder
355.0	1.0-1.5	0.40-0.6	0.50 6/	0.25	4.5-5.5	0.8 6/	0.35	0.25	0.0008	-	-	-	0.05	0.15	Remainder
356.0	0.25	0.20-0.40	0.35	-	6.5-7.5	0.60	0.35	0.25	0.0008	-	-	-	0.05	0.15	Remainder
D712.0	0.25	0.50-0.65	0.10	0.40-0.6	0.30	0.50	5.0-6.5	0.15-0.25	0.0008	-	-	-	0.05	0.20	Remainder

1/ All percentages are maximum unless shown as a range or otherwise indicated.
2/ Silicon plus iron shall not exceed 1.0 percent.
3/ Silicon plus iron shall not exceed 0.40 percent.
4/ Silicon plus iron shall not exceed 0.45 percent.
5/ Silicon plus iron shall not exceed 0.50 percent.
6/ If iron exceeds 0.45 percent, manganese should be present in an amount equal to one half the iron.
7/ Aluminum contents is the difference between 100.00 percent and the sum of all other metallic elements present in the amount of 0.010 percent or more each, expressed to the second decimal place.

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3.3.3 Weight. Welding rods and electrodes shall be furnished in the weights shown in table II, as specified (see 6.2). The net weight of the rods or electrodes shall not vary from the specified weight by more than ±10 percent.

TABLE II. Form, Standard Size and Weight

	Form		
	Straight lengths	Coiled	Spooled
Specified Nominal Diameter - inches	1/16(0.062), 3/32(0.094) 1/8(0.125), 5/32(0.156) 3/16(0.188), 1/4(0.250)	1/16(0.062), 3/32(0.094) 1/8(0.125), 5/32(0.156) 3/16(0.188), 1/4(0.250)	0.020, 0.030 3/64(0.047), 1/16(0.062) 3/32(0.094), 1/8(0.125)
Weight - pounds	5-, 10-, 25-, 50-	25-, 50-, 65-	1-, 10-, 12-1/2-, 15-

3.4 Tensile strength.

3.4.1 Welding rods. Unless otherwise specified, welding rods furnished in straight lengths or coils may be in any temper. Unless otherwise specified, welding rods furnished on spools shall conform to the requirements of table III for the tensile strength as applicable to the material specified (see 6.2).

3.4.2 Welding electrodes. All welding electrodes shall conform to the requirements of table III for the tensile strength as applicable to the material specified.

TABLE III. Minimum Tensile Strength of Spooled Welding Rods and Electrodes

Class	Tensile Strength, Minimum, PSI	Class	Tensile Strength, Minimum, PSI
1100	25,000	5039	40,000
1260	25,000	5183	42,000
2319	37,000 1/	5356	42,000
4043	27,000	5554	30,000
4047	30,000	5556	42,000
4145	27,000	5654	30,000

1/ Material with a diameter of 1/8 inch shall have a minimum tensile strength of 35,000 PSI.

3.5 Welding characteristics. When melted, using argon, helium or a mixture of the inert gases for shielding under a tungsten arc in gas tungsten-arc welding (GTAW), or as the electrode in gas metal-arc welding (GMAW); or under a gas flame such as oxyacetylene with a suitable flux conforming to MIL-F-6939, the welding rods and electrodes shall be capable of operating smoothly without interruption or

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excessive spatter, flow freely and uniformly without sputtering and other irregularities and produce a pool of molten weld metal of uniform appearance and form. The weld bead produced shall be of uniform appearance. The weld deposit shall not show rejectable defects such as porosity, cavities, cracks, discontinuity, undercutting, etc., when examined by penetrant inspection and radiography.

3.6 Identification. The class designation shall be stamped or molded at least once on each straight length rod near one end. In lieu of stamping by incise marking rolls or by imprinting, rods may be identified by tagging each 5 or 10 pounds of a particular class and diameter. A metal tag, stamped or embossed with class and diameter designation shall be securely wired to each coil. The spool used for rod or electrode shall conform to MIL-W-10430. The class, diameter and net weight shall appear on at least one flange of all spools. Such identification shall be placed in a manner that is not readily removable. Unless otherwise specified, the identification shall be placed on the flange of 1-pound spools so as to be visible when the rod or electrode is payed-off in the counterclockwise direction. Unless otherwise specified, the identification shall be placed on the flange of 10-, 12 1/2- and 15-pound spools so as to be visible when the rod or electrode is payed-off in the clockwise direction (see 6.2).

3.7 Finish. Rods and electrodes shall be bare and have clean surfaces with no stains, discolorations, dirt, lubricants or other surface contamination. They shall be free from splices, slivers, laps, cracks, pipes and segregation, pitting, inclusions or other irregularities which would affect the welding characteristics, the operation of welding equipment or the properties of the deposited weld metal.

3.8 Workmanship. Rods and electrode shall be uniform in composition, color, quality, size, true to form, smooth, and free from injurious defects. Rods, furnished in cut lengths, shall be straight. Cast rods may be chemically cleaned, if necessary to obtain a clean surface. Rod and electrode furnished in coils shall be free from kinks. Spooled rod and electrode shall be continuous and closely wound in layers in such a manner that kinks, waves and sharp bends are avoided. Spooled rod and electrode shall be free to unwind freely without restrictions caused by overlapping, wedging and tangling. Adjacent turns within a layer need not be touching. The outer end of coiled or spooled rod and electrode shall be readily located.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the

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inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.2 Lot. A lot shall consist of welding rods or electrodes of the same class, size and temper, produced under the same conditions and subjected to inspection at one time. Unless otherwise specified (see 6.2), the maximum weight of a lot shall be 6000 pounds.

4.3 Sampling. Unless otherwise specified, sampling plans and procedures for inspection shall be in accordance with the provisions set forth in MIL-STD-105. Samples shall be selected from either filled unit containers or from the production line immediately prior to packaging (after coiling or spooling). If selected from the production line, the total sample shall be selected throughout the run, so all parts of the run are represented.

4.3.1 For filled containers. Samples for examination of filled containers, both shipping containers filled with unit containers and filled unit containers, shall be selected at random from each lot in accordance with MIL-STD-105, Inspection Level II, Acceptable Quality Level 2.5 percent defective.

4.3.2 For visual and dimensional examination. Samples for visual and dimensional examination shall be selected at random from each lot in accordance with MIL-STD-105, Inspection Level II, Acceptable Quality Level of 2.5 percent defective for straight length rods and an Acceptable Quality Level of 2.5 percent defects for 100 feet of coiled or spooled material. For the coiled or spooled rod or electrode, a sufficient number of coils or spools shall be selected from each lot so that the sample footage for inspection shall consist of approximately 50 feet of material taken from the outer end of each sample coil or spool. The material selected in this sampling shall be taken from packages selected in the packaging inspection sampling (see 4.3.1) or from the production line (see 4.3).

4.3.3 For chemical analysis.

4.3.3.1 Ingot sampling. At least one sample shall be taken from each group of ingots of the same alloy poured simultaneously from the same source of molten metal by the producer for chemical analysis in accordance with 4.5.2. Ingots not conforming to the requirements of this specification shall be rejected. Complete ingot analysis records shall be available to the procuring activity at the producer's plant.

4.3.3.2 Finished product sampling. When compliance with 4.3.3.1 cannot be established, a random sample shall be selected from each lot as detailed in 4.3.2 in accordance with MIL-STD-105. Inspection Level S-1, acceptance number of zero for chemical analysis in accordance with 4.5.2.

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4.3.3.3 Preparation. Samples for chemical analysis shall be prepared in accordance with method 111 or method 112 of Fed. Test Method Std. No. 151. The samples shall be free from dirt, oil, grit and other foreign matter.

4.3.4 For tensile tests. At least two samples shall be selected at random from each lot of electrodes or spooled rods or from the production line as detailed in 4.3 for tensile testing in accordance with 4.5.3.

4.3.5 For welding characteristics. A random sample shall be selected from each lot of straight length or coiled welding rods, as detailed in 4.3.2, in accordance with MIL-STD-105, Inspection Level S-1, acceptance number of zero for the flat position weld bead test as detailed in 4.5.4 for determination of welding characteristics. For electrodes and spooled rods, one sample shall be selected for each 2000 pounds or fraction thereof in a lot, as detailed in 4.5.4 for the determination of the welding characteristics.

4.4 Quality conformance examination.

4.4.1 Containers. Each of the filled shipping containers and each unit package, selected in accordance with 4.3.1, shall be examined for construction defects, unsatisfactory markings, closure and weight to determine conformance to the requirements of section 5. Each filled unit container of rods or electrodes shall be weighed to determine net weight of contents for conformance to 3.3.3 and table II.

4.4.2 Visual examination. Sample rods and electrodes, selected in accordance with 4.3.2, shall be visually examined for conformance with the requirements of 3.3 for form, of 3.6 for identification, of 3.7 for finish and 3.8 for workmanship.

4.5 Quality conformance tests.

4.5.1 Dimensional measurements. The sample rods or electrodes selected in accordance with 4.3.2, shall be measured for length to determine conformance to 3.3.2 and for diameter to determine conformance to 3.3.1 and table II. The diameter shall be measured using standard micrometer methods or go-no-go gages.

4.5.2 Chemical composition. The sample selected from the ingot in accordance with 4.3.3.1 or the sample rods or electrodes selected in accordance with 4.3.3.2 shall be tested in accordance with method 111 or method 112 of Fed. Test Method Std. No. 151 to determine conformance to 3.2 and table I.

4.5.3 Tensile tests. The tensile strength of the full cross-section sampled spooled rods or electrodes, selected in accordance with 4.3.4, shall be tested in accordance with ASTM E8 to determine conformance to 3.4 and table III.

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4.5.4 Welding characteristics. To determine welding characteristics and usability, the welding rods and electrodes shall be tested by a weld bead depotition on the surface of a test panel in the flat position, unless welding of a jointed assembly is specified in the contract or order (see 6.2).

4.5.4.1 Equipment. For gas metal-arc and gas tungsten-arc welding, manual, semi-automatic or automatic equipment with single operator power sources shall be used. For gas welding standard torches, hoses and regulators suitable for use with oxyacetylene or oxyhydrogen welding shall be used. Equipment shall be operated by an experienced welder or welding operator.

4.5.4.2 Specimen panel materials. Sheet and plate materials used for specimens shall be in accordance with table IV. For the weld bead test, the panel shall be of sufficient length and width to allow an overlay bead not less than 12 inches long, to be centered on the test panel. The thickness of the panel shall be suitable for use with the sample rods or electrodes selected in accordance with 4.3.5. For the welded assembly test, test plates shall be prepared in accordance with figure 1 and the notes thereof. Panels shall be thoroughly cleaned to remove oil, grease and thick oxide film. Oil and grease may be removed by dipping or wiping the test plates with a suitable solvent. Oxide film may be removed mechanically by wire brushing with a stainless steel brush, scraping or filing, or by chemical cleaning.

4.5.4.3 Welding operations. The supplier may use any welding technique considered best suited for satisfactory weld deposit. For the gas tungsten-arc process, alternating current is recommended to be used. Straight polarity direct current may be used for automatic gas tungsten-arc welding setups but should not be used for manual welding. For the gas metal-arc process, reverse polarity direct current shall be used. Settings shall provide for stable arc characteristics, suitable wetting and adequate penetration. The welding may be conducted in an inert gas welding chamber capable of maintaining contaminant free atmosphere of helium, BB-H-1168, or argon, MIL-A-18455, having maximum dew point of -75°C. If welding is performed outside a chamber with GTAW or GMAW, the work shall be shielded by argon, helium, or a mixture of these two inert gases. The supplier should recommend the exact gas composition for welding. For OAW or OHW processes, a flux conforming to MIL-F-6939 shall be employed. The weld bead test shall be made by laying a weld bead on the surface of the test panel in the flat position with the sample rod or electrode selected in accordance with 4.3.5. Straight lengths and coiled rods shall be tested only with the flat position procedure using GTAW, OAW or OHW processes. Spooled rods and electrodes both spooled and coiled, shall be tested with the flat position weld deposition using GTAW or GMAW processes, unless the welding of a butt joint assembly is specified (see 4.5.4). The butt joint weld test shall be performed only by GMAW with the test assembly positioned as stated in the notes of figure 1. The selected sample rods or electrodes shall be used for making welds by stringer beading. For multiple pass welds, the layer thickness shall not exceed 3/16 inch. Interpass temperature shall not exceed 150°F. Water quenching between passes shall not be permitted. No preheat or post-heat treatments shall be used.

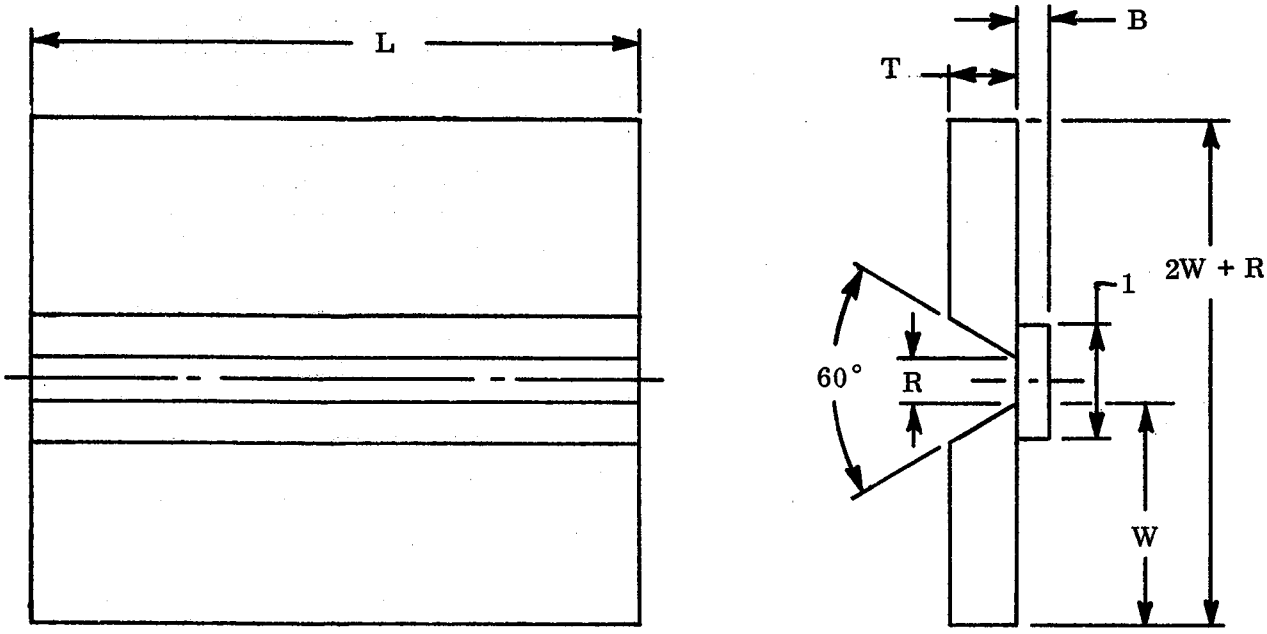
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TABLE IV. Plate Materials for Welding Tests

Filler Metal - Class	Plate Material - Alloy (conforming to)
1100 and 1260	1100 (QQ-A-250/1), 3003 (QQ-A-250/2) or 1060 (AMS-4000)
2319	3003 (QQ-A-250/2) or 2219 (MIL-A-8920)
4043 and 4047	3003 (QQ-A-250/2) or 6061 (QQ-A-250/11)
4145	2219 (MIL-A-8920) or 6061 (QQ-A-250/11)
5039	5083 (QQ-A-250/6), 5086 (QQ-A-250/7) or 5456 (QQ-A-250/9)
5554 and 5654	5052 (QQ-A-250/8) or 5454 (QQ-A-250/10)
5183, 5356, 5556	5083 (QQ-A-250/6), 5086 (QQ-A-250/7) or 5456 (QQ-A-250/9)
295.0	295.0 (QQ-A-601)
242.0	242.0 (AMS-4222)
355.0	355.0 (QQ-A-596)
356.0	356.0 (QQ-A-596) or (QQ-A-601)
D 712.0	D 712.0 (QQ-A-601)

4.5.4.4 Evaluation. The operating characteristics of the welding rods or electrodes shall be observed during fabrication of the weld bead or joint as well as feed performance for conformance with 3.5, 3.7 and 3.8. The surface of weld bead or joint, and adjacent base metal shall be examined visually. The weld area shall be smooth, uniform, free from rejectable surface porosity, cracking, excessive spatter, undercutting, overlapping and other defects to be in conformance with 3.5. The surface bead test plate shall be penetrant inspected in accordance with MIL-I-6866, using either type I or type II, method C material (solvent removable) for surface discontinuity, undercutting, cracks, porosity and other defects to determine conformance with 3.5. Limitations detailed in MIL-STD-437 for porosity and other defects shall be used as the inspection standards for the flat position weld. The welded butt joint assembly shall be inspected in accordance with MIL-STD-453, after removal of the weld reinforcement and the backing strip, to determine the extent of porosity and other defects for conformance with 3.5. The criteria for radiographic defects shall be as detailed in MIL-STD-437.

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Test plate for checking welding characteristics and soundness of weld deposit.

Backing strip shall be of same material as plate. Back strip may be tack welded or clamped to assembly.

Dimensions in inches

Dimensions	Filler Material - diameters		
	0.030 to 3/64	1/16 to 3/32	1/8
Plate - Length (L)	10 min.	10 min.	10 min.
Plate - Width (W)	5 min.	5 min.	5 min.
Plate - Thickness (T)	1/4	3/8	3/8
Backing strip (B)	1/4	3/8	3/8
Root opening (R)	1/4	3/8	1/2
Welding position	Overhead	Overhead	Flat

FIGURE 1. Plates for Butt Weld Test.

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4.6 Rejection and retests. A sample package of rods, coil of wire, or sample spool which contains material not conforming to the requirements of 4.4.2 shall be counted as defective. The number of defective rods, defective feet of coiled wire or the number of defective spools shall not exceed the acceptance number. All tests shall be performed with the material after the last procuring operation. If the results of the inspection penetrant test or of the radiographic examination are determined to be unsatisfactory, two retests shall be permitted. The results of both retests shall be satisfactory for lot acceptance.

5. PREPARATION FOR DELIVERY

5.1 Packaging. Packaging shall be level A or C, as specified (see 6.2).

5.1.1 Level A.

5.1.1.1 Rods. Unless otherwise specified in the contract or order, rods shall be packaged in accordance with MIL-W-10430. A unit package shall contain one class and size as specified. Rods in straight lengths shall be unit packaged in 5-, 10-, 25-, or 50-pound quantities, net weight, as specified. Unit containers shall be class 6a, 6b, 6c or 7 of MIL-W-10430. When specified in the contract or order, rods shall be wrapped as detailed in MIL-W-10430 for classes 3a and 3b containers prior to innerpackaging.

5.1.1.2 Coils. Unless otherwise specified in the contract or order, welding rods or electrodes as coils shall be packaged in accordance with MIL-W-10430. A unit package shall contain one continuous length of the class and size specified. Coils shall be unit packaged in 25-, 50- or 65-pound quantities, net weight, as specified. The outside diameter (O.D.) and inside diameter (I.D.) of the coil shall also be specified. Unit containers shall be class 3d of MIL-W-10430.

5.1.1.3 Spools. Unless otherwise specified in the contract or order, welding rods or electrodes wound on spools shall be packaged in accordance with MIL-W-10430. A unit package shall contain one continuous length of the class and size specified. Rod or electrode wound on spools shall be unit packaged in 1-, 10-, 12 1/2- or 15-pound quantities, net weight, as specified. Unit containers shall be class 3a or 3b of MIL-W-10430.

5.1.2 Level C. Welding rods and electrodes shall be packaged in accordance with level C requirements of MIL-W-10430 to afford the minimum degree of protection necessary to prevent deterioration or damage during shipment under normal environmental conditions and commercial modes of transportation.

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5.2 Packing. Packing shall be level A, B, or C, as specified (see 6.2).

5.2.1 Level A. Welding rods and electrodes, packaged as specified, shall be packed for overseas shipment in accordance with MIL-W-10430.

5.2.2 Level B. Welding rods and electrodes, packaged as specified, shall be packed for domestic shipment in accordance with MIL-W-10430.

5.2.3 Level C. Welding rods and electrodes that require overpacking by the carrier shall be packed in accordance with MIL-W-10430 in exterior type shipping containers in a manner that will insure safe transportation at the lowest rate to the point of delivery and shall meet as a minimum, the requirements of the rules and regulations applicable to the mode of transportation selected.

5.3 Marking. In addition to any special markings required by the contract or order (see 6.2), interior packages, exterior shipping containers and palletized unit loads shall be marked in accordance with MIL-W-10430 and Fed. Std. No. 123 for Civil Agencies or MIL-STD-129 for Military Activities.

5.3.1 Unit containers. Unit containers shall bear a printed label in a conspicuous location on or near the manufacturer's label and on any special instructions which may be enclosed as follows:

WARNING

Welding may produce fumes and gases hazardous to health. Avoid breathing these fumes and gases. Use adequate ventilation. See American National Standard Z49.1, Safety in Welding and Cutting.

5.4 Shipping container tests. Tests on shipping containers as specified in MIL-W-10430 shall apply to levels A and B packing only.

6. NOTES

6.1 Intended use.

6.1.1 The filler rods and electrodes of aluminum and aluminum alloys, as shown in table V, are recommended as being suitable for use with arc welding processes; namely, the gas tungsten-arc (GTA) and the gas metal-arc (GMA). Rods, as applicable to the material detailed herein, refers to straightened-and-cut lengths of wire or cast lengths which are intended essentially for gas tungsten-arc (GTAW) or for gas (OAW or OHW) welding processes. Rods or electrodes, coiled or on spools, as detailed herein, are intended for use with automatic or semi-automatic welding machines employing continuous feed devices. Gas metal-arc welding (GMAW) uses a continuous feed wire as the electrode, whereas, the gas tungsten-arc (GTAW) may have the rod added manually or by means of feeding mechanism when machine welding is used. Only classes 1100, 1260, 4043, 4047 and 4145 are ordinarily used in gas

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welding processes (oxyacetylene or oxyhydrogen). Classes 2319 and 4145 are intended for welding the copper type aluminum alloys (the 2xxx group).

6.1.2 The correlation between the welding tests (see 4.5.4), used in this specification, and acceptable welding processes are as follows:

Material	Type of Test Panel (see 4.5.4.2)	Welded by	Acceptable Welding Processes
Rod or electrode	Butt	GMAW	All (GMAW, GTAW, OAH, OHW, etc.)
Rod or electrode	Flat	GMAW	GMAW, GTAW, OAW, OHW
Rod	Flat	GTAW	GTAW, OAW, OHW
Rod	Flat	OAW or OHW	OAW, OHW

6.1.3 For foundry repair welding or assembly welding of sand and permanent-mold castings of heat treatable aluminum alloys, cast filler metal should be of the same alloy as the casting being welded. Table V lists classes of wrought filler alloys commonly used for welding the casting alloys to other casting alloys and various wrought alloys. In addition the following classes of cast welding rods, listed in table I, may be used with the following casting alloys.

Class - Cast Rods		For Casting Alloys	
New Designation	Former Designations	New Designation	Former Designations
295.0	C4A or FS-RAl 195	295.0, B 295.0	195, B 195
242.0	CN42A or FS-RAl 142	242.0, A 242.0	142, A 142
355.0	SC51A or FS-RAl 355	355.0, C 355.0 B 443.0	355, C 355, 43
356.0	SG70A or FS-RAl 356	356.0, A 356.0 A 440	356, A 356, 344
D 712.0	2G61A or FS-RAl 40E	D 712.0, A 712.0 C 712.0	40E, D 612, A 612, C 612

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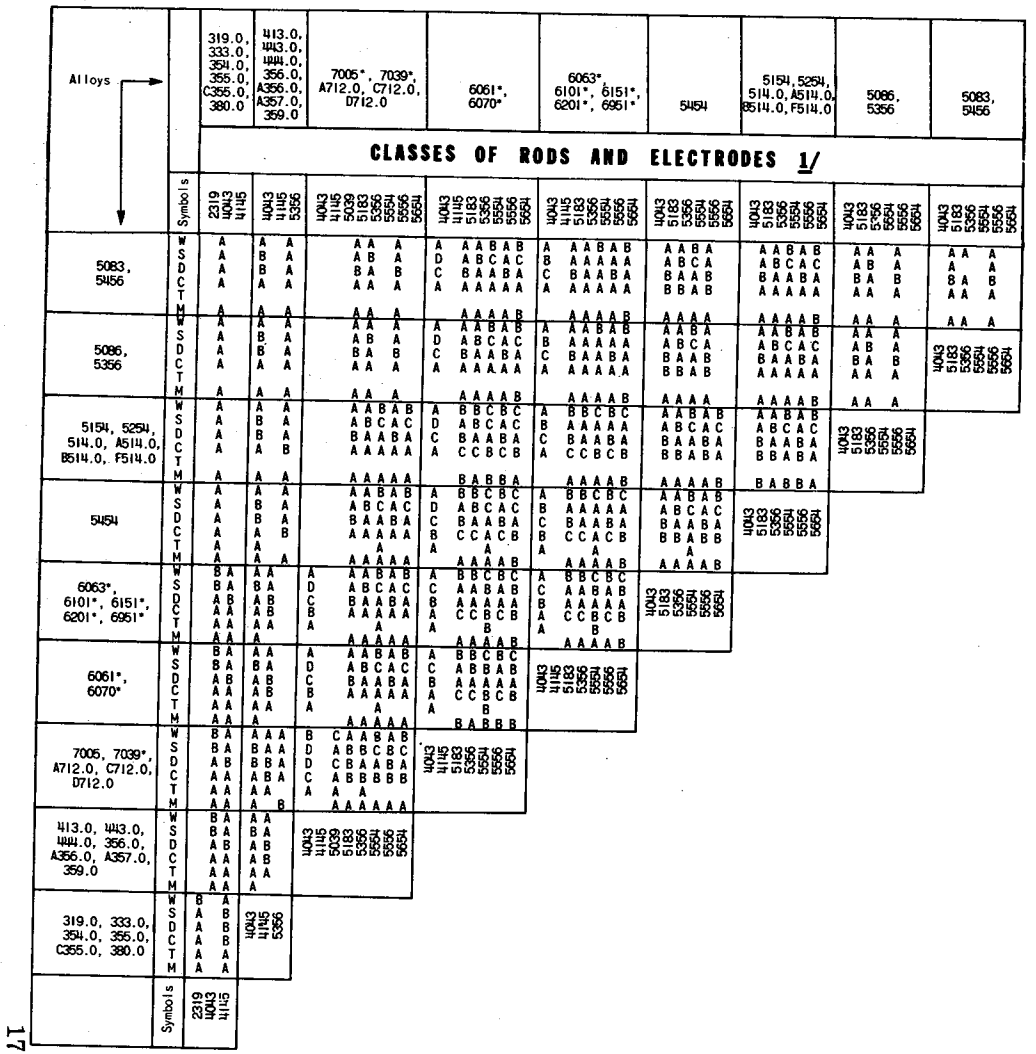
TABLE V GUIDE TO THE CHOICE OF RODS AND ELECTRODES FOR WELDING

CLASSES OF RODS AND ELECTRODES 1/

Alloys	Symbols	319.0, 333.0, 354.0, 355.0, C355.0, 360.0	413.0, 443.0, 444.0, 356.0, A356.0, 357.0, 359.0	7005*, 7039*, A712.0, C712.0, 0712.0	6061*, 6070*	6063*, 6101*, 6151*, 6201*, 6861*	5454	5154, 5254, 514.0, A514.0, F514.0, 8214.0, F514.0	5086, 5356	5083, 5456	5652, 5652	5005, 5050	Alc. 3004	3004	3003, Alc. 3003	2219*	2014*	1100	1060, EC
1060, EC	W S D C T M	A A A A A A	A A A A A A	A A A A A A	AABB B AAAA A CDBA B AB AA AA A	AABB B AAAA A CDBA B AB AA AA A	ABBCB BAAAA CBAAB CBBAB A AAAA	ABBCBC BAAAAA CBAABA CBBABA AAAAB	AAA A BAA A CBA B BAA A AA A	AAA A BAA A CBA B BAA A AA A	ABB B BAA B CBA B A AA A	A A A A	CABCCC CBBAAA ACDBBB AABCCC AAA A BBB	A BBCB D ABCA D CBAC A CCBC A A AAAA	BA AB AA AA	CABCCC CBBAA ACDBB AABCC AAA A BB	BA AB AA AA	BCA BCA AAB AAA BA
1100	W S D C T M	BA AA AB AA AA AA	BA AA AB AA AA AA	BA AA AB AA AA AA	AABB B AAAA A CDBA B AB AA AA A	AABB B AAAA A CDBA B AB AA AA A	ABBCB BAAAA CBAAB CBBAB A AAAA	ABBCBC BAAAAA CBAABA CBBABA AAAAB	AAA A BAA A CBA B BAA A AA A	AAA A BAA A CBA B BAA A AA A	ABB B BAA B CBA B A AA A	A A A A	CABCCC CBBAAA ACDBBB AABCCC AAA A BBB	A BBCB D ABCA D CBAC A CCBC A A AAAA	BA AB AA AA	CABCC CBBA ACDBB AABCC AAA A BB	BA AB AA AA	BCA BCA AAB AAA BA
2014*	W S D C T M	BCA ACB ABC AAA AAA AAA	BA AB AB AA AA AA	BA AB AB AA AA AA	AABB B AAAA A CDBA B AB AA AA A	AABB B AAAA A CDBA B AB AA AA A	ABBCB BAAAA CBAAB CBBAB A AAAA	ABBCBC BAAAAA CBAABA CBBABA AAAAB	AAA A BAA A CBA B BAA A AA A	AAA A BAA A CBA B BAA A AA A	ABB B BAA B CBA B A AA A	A A A A	CABCCC CBBAAA ACDBBB AABCCC AAA A BBB	A BBCB D ABCA D CBAC A CCBC A A AAAA	BA AB AA AA	CABCC CBBA ACDBB AABCC AAA A BB	BA AB AA AA	BCA BCA AAB AAA BA
2219*	W S D C T M	BCA ACB ABC AAA AAA AAA	BA AA AB AA AA AA	BA AA AB AA AA AA	AABB B AAAA A CDBA B AB AA AA A	AABB B AAAA A CDBA B AB AA AA A	ABBCB BAAAA CBAAB CBBAB A AAAA	ABBCBC BAAAAA CBAABA CBBABA AAAAB	AAA A BAA A CBA B BAA A AA A	AAA A BAA A CBA B BAA A AA A	ABB B BAA B CBA B A AA A	A A A A	CABCCC CBBAAA ACDBBB AABCCC AAA A BBB	A BBCB D ABCA D CBAC A CCBC A A AAAA	BA AB AA AA	CABCC CBBA ACDBB AABCC AAA A BB	BA AB AA AA	BCA BCA AAB AAA BA
3003, Alc. 3003	W S D C T M	A BA AB AA AA AA	BB B AA A BA B AA A AA A AA A	A BA AB AA AA AA	AABB B BAAA A CDBA B AB AA AA A	AABB B BAAA A CDBA B AB AA AA A	ABBCB BAAAA CBAAB CBBAB A AAAA	ABBCBC BAAAAA CBAABA CBBABA AAAAB	AAA A BAA A CBA B BAA A AA A	AAA A BAA A CBA B BAA A AA A	ABB B BAA B CBA B A AA A	A A A A	CABCCC CBBAAA ACDBBB AABCCC AAA A BBB	A BBCB D ABCA D CBAC A CCBC A A AAAA	BA AB AA AA	CABCC CBBA ACDBB AABCC AAA A BB	BA AB AA AA	BCA BCA AAB AAA BA
3004	W S D C T M	A BA AA AA AA AA	BB B AA A BA B AA A AA A AA A	A BA AB AA AA AA	AABB B BAAA A CDBA B AB AA AA A	AABB B BAAA A CDBA B AB AA AA A	ABBCB BAAAA CBAAB CBBAB A AAAA	ABBCBC BAAAAA CBAABA CBBABA AAAAB	AAA A BAA A CBA B BAA A AA A	AAA A BAA A CBA B BAA A AA A	ABB B BAA B CBA B A AA A	A A A A	CABCCC CBBAAA ACDBBB AABCCC AAA A BBB	A BBCB D ABCA D CBAC A CCBC A A AAAA	BA AB AA AA	CABCC CBBA ACDBB AABCC AAA A BB	BA AB AA AA	BCA BCA AAB AAA BA
Alc. 3004	W S D C T M	A B A B A A	BB B AA A BA B AA A AA A AA A	A BA AB AA AA AA	AABB B BAAA A CDBA B AB AA AA A	AABB B BAAA A CDBA B AB AA AA A	ABBCB BAAAA CBAAB CBBAB A AAAA	ABBCBC BAAAAA CBAABA CBBABA AAAAB	AAA A BAA A CBA B BAA A AA A	AAA A BAA A CBA B BAA A AA A	ABB B BAA B CBA B A AA A	A A A A	CABCCC CBBAAA ACDBBB AABCCC AAA A BBB	A BBCB D ABCA D CBAC A CCBC A A AAAA	BA AB AA AA	CABCC CBBA ACDBB AABCC AAA A BB	BA AB AA AA	BCA BCA AAB AAA BA
5005, 5050	W S D C T M	A B A B A A	BB B AA A BA B AA A AA A AA A	A BA AB AA AA AA	AABB B BAAA A CDBA B AB AA AA A	AABB B BAAA A CDBA B AB AA AA A	ABBCB BAAAA CBAAB CBBAB A AAAA	ABBCBC BAAAAA CBAABA CBBABA AAAAB	AAA A BAA A CBA B BAA A AA A	AAA A BAA A CBA B BAA A AA A	ABB B BAA B CBA B A AA A	A A A A	CABCCC CBBAAA ACDBBB AABCCC AAA A BBB	A BBCB D ABCA D CBAC A CCBC A A AAAA	BA AB AA AA	CABCC CBBA ACDBB AABCC AAA A BB	BA AB AA AA	BCA BCA AAB AAA BA
5052, 5652	W S D C T M	A B A B A A	BB B AA A BA B AA A AA A AA A	A BA AB AA AA AA	AABB B BAAA A CDBA B AB AA AA A	AABB B BAAA A CDBA B AB AA AA A	ABBCB BAAAA CBAAB CBBAB A AAAA	ABBCBC BAAAAA CBAABA CBBABA AAAAB	AAA A BAA A CBA B BAA A AA A	AAA A BAA A CBA B BAA A AA A	ABB B BAA B CBA B A AA A	A A A A	CABCCC CBBAAA ACDBBB AABCCC AAA A BBB	A BBCB D ABCA D CBAC A CCBC A A AAAA	BA AB AA AA	CABCC CBBA ACDBB AABCC AAA A BB	BA AB AA AA	BCA BCA AAB AAA BA

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TABLE V GUIDE TO THE CHOICE OF RODS AND ELECTRODES FOR WELDING (Cont.)



*Rating does not cover these alloys when heat treated after welding.

1/ Class 5183 may be substituted for 5556, except when welding base metals 5456 to 5456. Welds with 5556 are slightly stronger, whereas welds with 5183 are slightly more ductile.

LEGEND

Rods and electrodes are rated on the following characteristics:

Symbol	Characteristic
W	Ease of welding.
S	Strength of welded joint ("as-welded" condition). (Rating applies particularly to fillet welds. All rods and electrodes rated will develop presently specified minimum strengths for butt welds.)
D	Ductility. (Rating is based upon free bend elongation of the weld.)
C	Corrosion resistance in continuous or alternate immersion in fresh or salt water.
T	Suitability for service at sustained temperatures above 150°F.
M	Color match after anodizing.

A, B, C and D are relative ratings in decreasing order of merit. The ratings have relative meaning only within a given block. Combinations having no rating are not usually recommended.

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6.1.4 Classes 4145 and 4047 filler materials, specified herein, are the equivalents of class FS-B Al Si-3 and FS-B Al Si-4, respectively, that comply with the requirements of QQ-B-655, Brazing Alloys, Aluminum and Magnesium Filler Metals, when used for brazing in the form of wire and rod.

6.1.5 The filler materials in this document are not considered suitable for processes where flux-coated electrodes rather than bare electrodes are required, such as the shielded metal-arc, shielded carbon-arc or the atomic hydrogen. Such filler materials shall be in accordance with MIL-E-15597.

6.2 Ordering data. Purchasers should select the preferred options permitted herein and include the following information in procurement documents:

a. Title, number, and date of this document

b. Alloy class or composition (see 1.2, 3.2 and table I)

c. Rods, if applicable, furnished as drawn or cast (see 3.1.1)

d. Form, as rods or electrode, if applicable, in coil or wound on spools (see 3.3)

e. Size, nominal diameter (see 3.3.1 and table II)

f. Length of drawn rod, if other than 36 inches long. Length of cast rod (see 3.3.2)

g. Weight of rods or electrode (see 3.3.3, table II, 5.1.1.1, 5.1.1.2 and 5.1.1.3)

h. Temper of wrought rods and electrodes, if applicable (see 3.4 and 3.4.1)

i. Spool identification, if other than specified (see 3.6)

j. Lot size, if other than specified (see 4.2)

k. Sampling plans and procedures, if other than specified (see 4.3)

l. Butt weld test required on coiled electrodes or spooled rods and electrodes (see 4.5.4)

m. Whether levels A or C packaging is required (see 5.1)

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n. Whether levels A, B or C packing is required (see 5.2)

o. Additional markings, if necessary (see 5.3)

6.2.1 Welding rods and electrode shall be ordered by the pound. When ordering filler wire for use with automatic welding machines, the net weight, the coil dimensions and dimensions of any spools which may be required for use with the feed mechanism of the respective welding machines should be specified.

6.3 Definitions.

6.3.1 Welding rod. Welding rod is defined as round filler metal used for welding wherein the filler metal does not conduct electrical current.

6.3.2 Welding electrode. Welding electrode is defined as round filler metal used for welding wherein the filler metal is a component of the welding circuit through which electrical current is conducted.

6.4 Specification for Aluminum and Aluminum Alloy Welding Rods and Bare Electrodes, AWS A5.10, prepared by the American Welding Society, may be used as a guide and source for further information as to welding procedures and selection of filler metal for welding.

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NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

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